

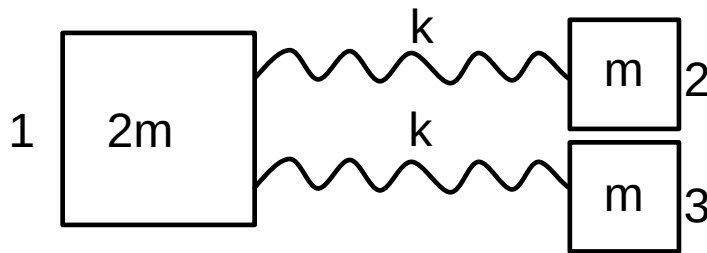


OSCILLATIONS, WAVES AND OPTICS

PHYSICS FOR CHEMISTRY STUDENTS

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REVIEW



$$m\ddot{x} = -kx(t)$$

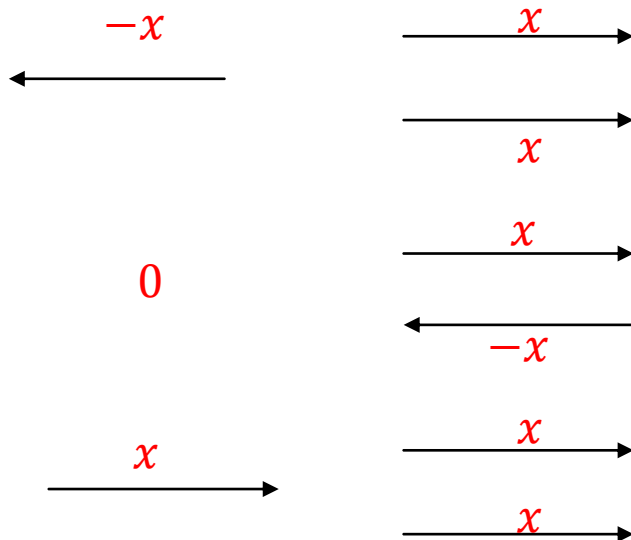
Normal Modes:

$$x_1(t) = -x_2(t) = -x_3(t) = a_0 \cos\left(\sqrt{\frac{2k}{m}}t + \phi\right)$$

(1)

(2)

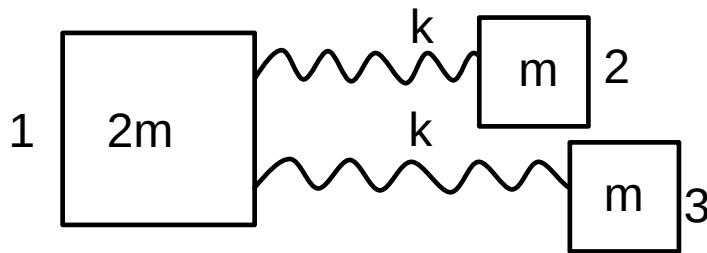
(3)



$$\begin{cases} x_1(t) = 0 \cdot \cos\left(\sqrt{\frac{k}{m}}t + \phi\right) \\ x_2(t) = -x_3(t) = b_0 \cos\left(\sqrt{\frac{k}{m}}t + \phi\right) \end{cases}$$

$$x_1(t) = x_2(t) = x_3(t) = vt + c$$

REVIEW



$$\ddot{\mathbf{X}} = \begin{pmatrix} \ddot{x}_1 \\ \ddot{x}_2 \\ \ddot{x}_3 \end{pmatrix}$$

$$\mathbf{X} = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}$$

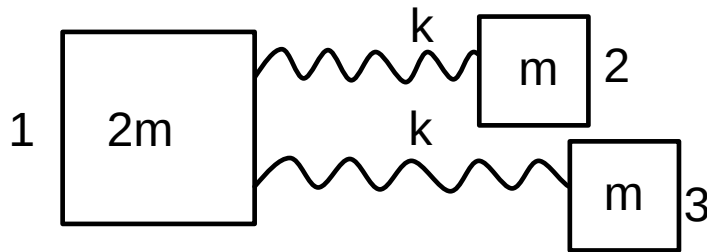
$$\mathbf{\Lambda} = \begin{pmatrix} 2k/2m & -k/2m & -k/2m \\ -k/m & k/m & 0 \\ -k/m & 0 & k/m \end{pmatrix}$$

$$\begin{cases} 2m\ddot{x}_1 = -k[x_1 - x_2] - k[x_1 - x_3] \\ m\ddot{x}_2 = -k[x_2 - x_1] \\ m\ddot{x}_3 = -k[x_3 - x_1] \end{cases}$$

$$\begin{cases} \ddot{x}_1 = -k/m x_1 + k/2m x_2 + k/2m x_3 \\ \ddot{x}_2 = k/m x_1 - k/m x_2 \\ \ddot{x}_3 = k/m x_1 - k/m x_3 \end{cases}$$

$$\ddot{\mathbf{X}} = -\mathbf{\Lambda}\mathbf{X}$$

REVIEW



$$\ddot{\mathbf{X}} = -\Lambda \mathbf{X}$$

Trial solution : $\mathbf{X}(t) = A \cos(\omega t + \phi)$

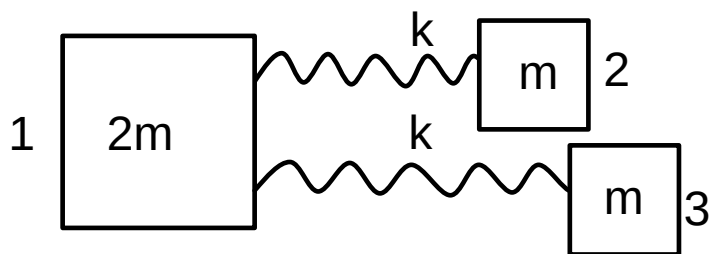
Characteristic eq.:

$$\Lambda \mathbf{A} - \omega^2 \mathbf{A} = \mathbf{0}$$

$$\det(\Lambda - \omega^2 \mathbf{I}) = 0$$

$$\det \begin{pmatrix} 2k/2m - \omega^2 & -k/2m & -k/2m \\ -k/m & k/m - \omega^2 & 0 \\ -k/m & 0 & k/m - \omega^2 \end{pmatrix} = 0$$

REVIEW



$$(\mathbf{\Lambda} - \omega^2 \mathbf{I})\mathbf{A} = \mathbf{0}$$

$$\begin{pmatrix} \omega_0^2 - \omega^2 & -1/2 \omega_0^2 & -1/2 \omega_0^2 \\ -\omega_0^2 & \omega_0^2 - \omega^2 & 0 \\ -\omega_0^2 & 0 & \omega_0^2 - \omega^2 \end{pmatrix} \mathbf{A} = \mathbf{0}$$

$$\omega^2 = \omega_0^2$$

$$\begin{pmatrix} 0 & -1/2 & -1/2 \\ -1 & 0 & 0 \\ -1 & 0 & 0 \end{pmatrix} \mathbf{A} = \mathbf{0}$$

$$\mathbf{A} = a \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix}$$

$$\omega^2 = 2\omega_0^2$$

$$\begin{pmatrix} 1 & 1/2 & 1/2 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix} \mathbf{A} = \mathbf{0}$$

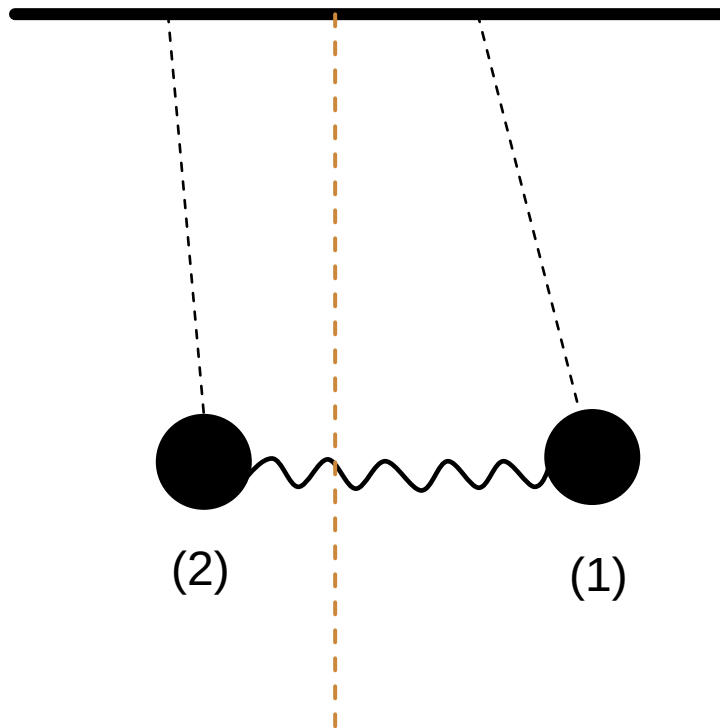
$$\mathbf{A} = b \begin{pmatrix} -1 \\ 1 \\ 1 \end{pmatrix}$$

$$\omega^2 = 0$$

$$\begin{pmatrix} 1 & -1/2 & -1/2 \\ -1 & 1 & 0 \\ -1 & 0 & 1 \end{pmatrix} \mathbf{A} = \mathbf{0}$$

$$\mathbf{A} = c \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$$

REVIEW

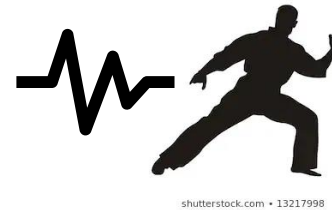


$$x_j(t) = x_{jc}(t) + b_j e^{i\omega_d t}$$

New equation of motion:

$$\ddot{\mathbf{X}} = -\mathbf{\Lambda}\mathbf{X} + \mathbf{F}\cos(\omega_d t)$$

$$(\mathbf{\Lambda} - \omega_d^2 \mathbf{I}) \mathbf{B} = \mathbf{F}$$



$$f_{dr} = f_0 \cos(\omega_d t)$$

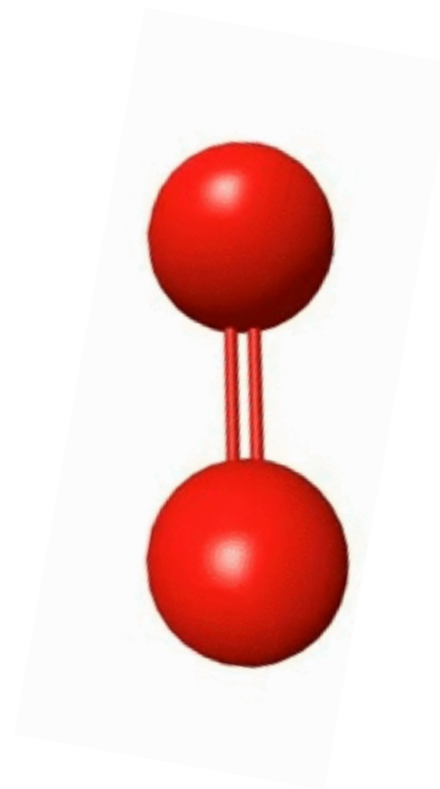
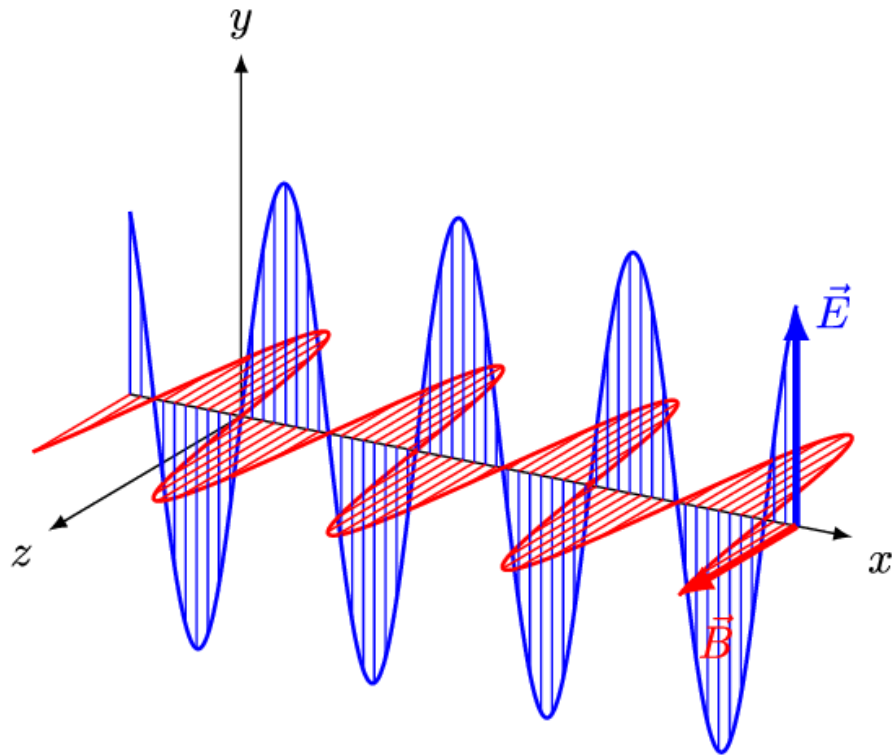
$$\omega_d^2 \rightarrow \omega_1^2$$

$$\frac{b_1}{b_2} \rightarrow 1$$

$$\omega_d^2 \rightarrow \omega_2^2$$

$$\frac{b_1}{b_2} \rightarrow -1$$

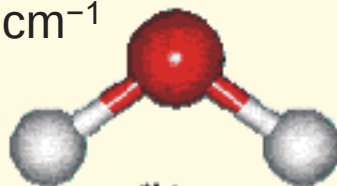
REVIEW



REVIEW

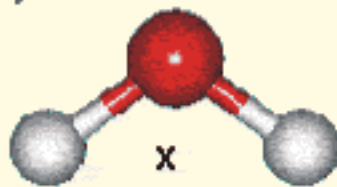
H₂O Normal modes:
3N-3

3657 cm⁻¹



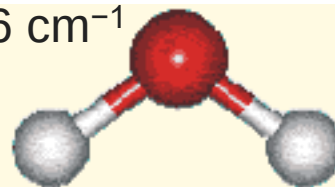
ν_1

symmetric stretch



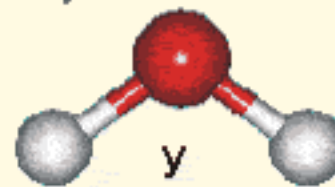
x

3756 cm⁻¹



ν_3

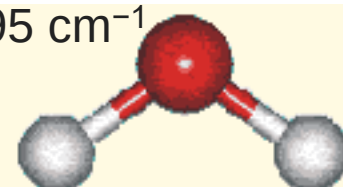
asymmetric stretch



y

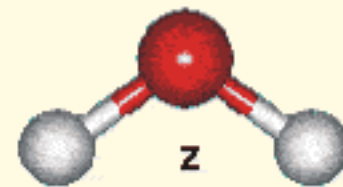
librations

1595 cm⁻¹

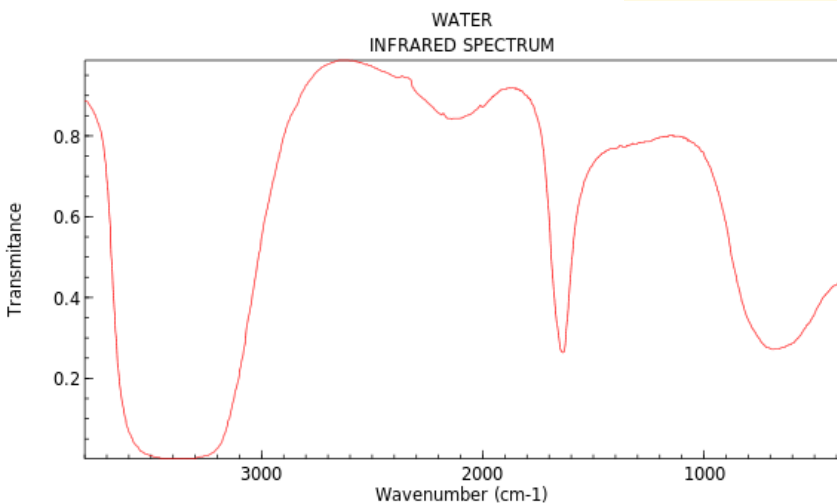


ν_2

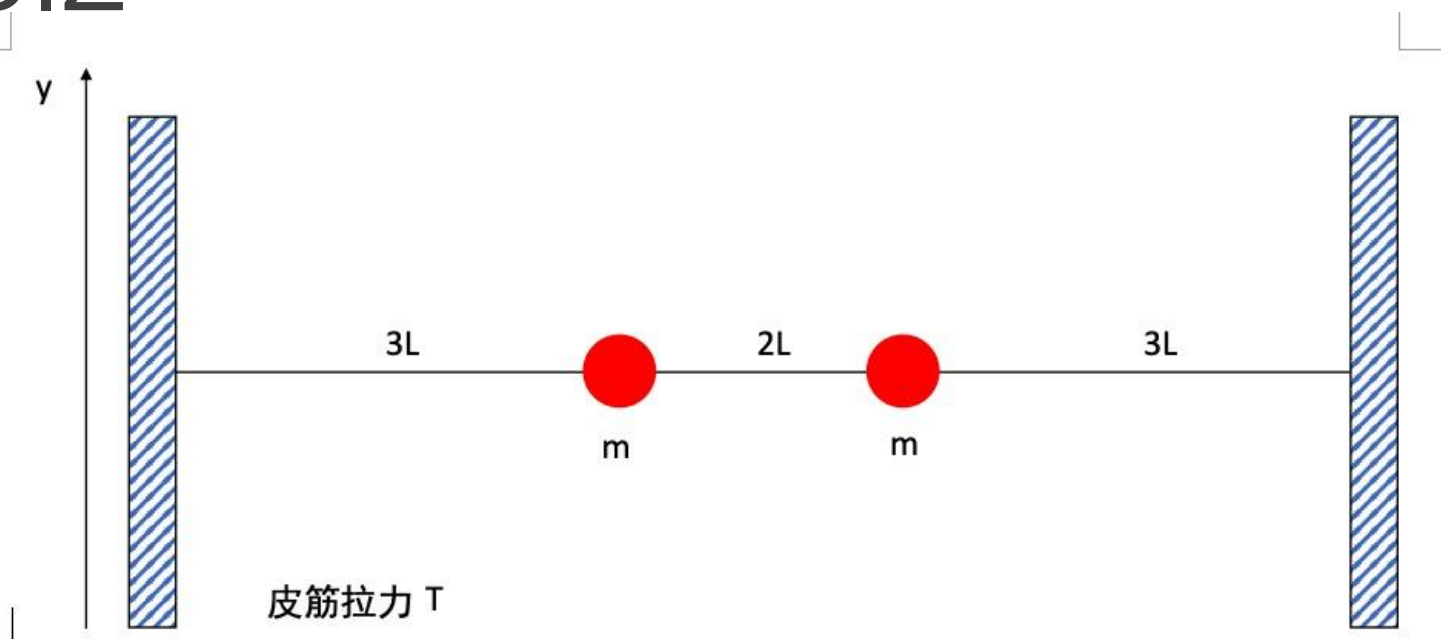
bend



z



QUIZ



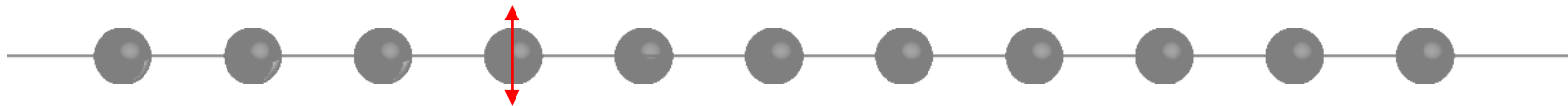
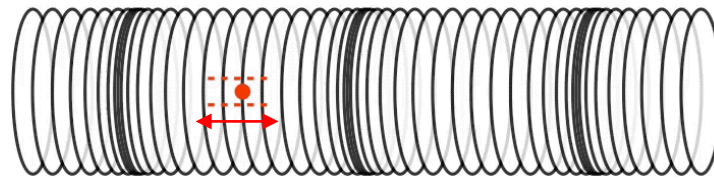
三根皮筋牵引 2 个质量为 m 的物体（小球），皮筋拉力为 T ，两端皮筋分别固定在墙面。假设小球只在 y 方向运动，除了拉力之外，不考虑任何其它作用力。

1. 假设小球沿 y 方向做小幅振动（皮筋拉力保持不变），给出该体系的简正模的频率。
2. 初始时刻其中一个小球处于平衡位置，另外一个小球在 y_0 位置，初始时刻两个小球的速度均为 0。写出小球位置随时间变化的表达式。

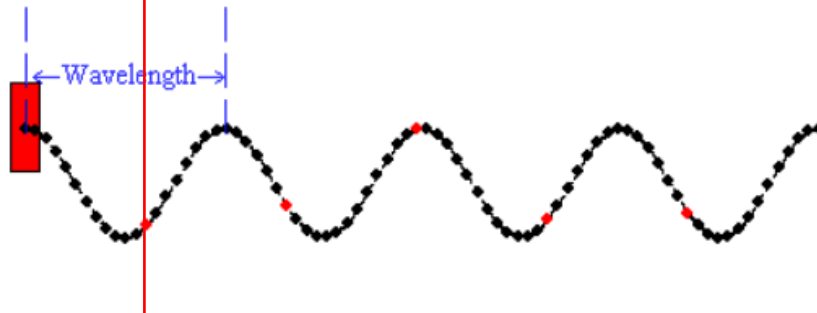
CHAPTER 3. WAVE



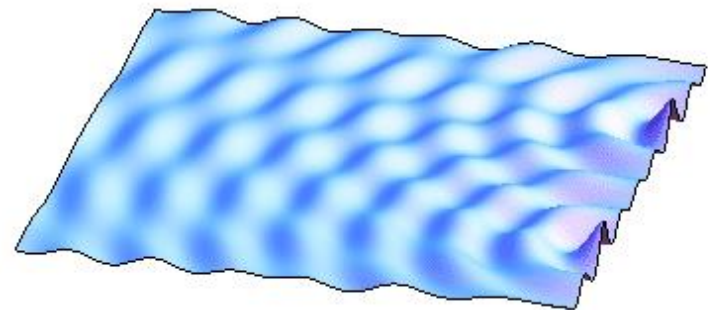
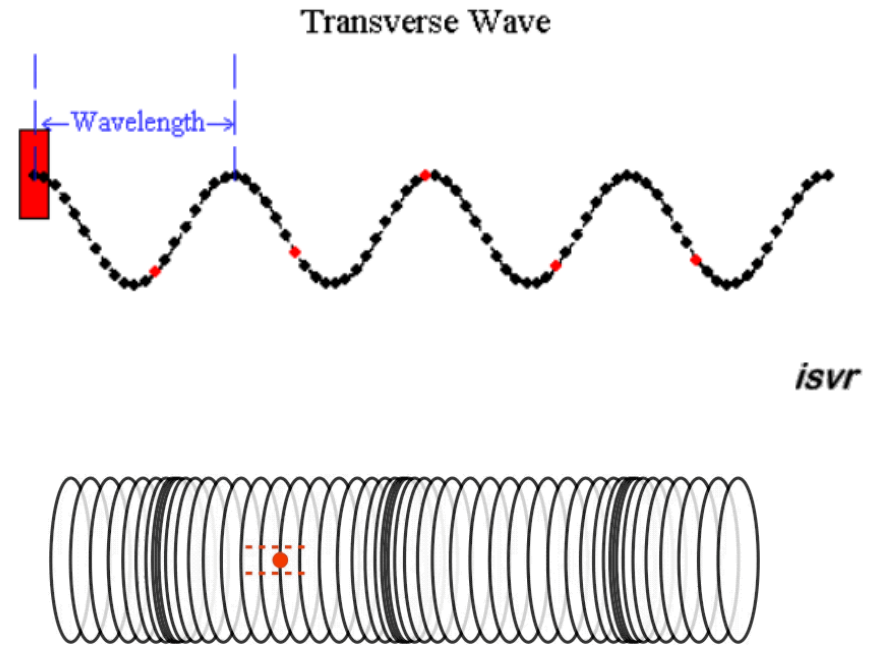
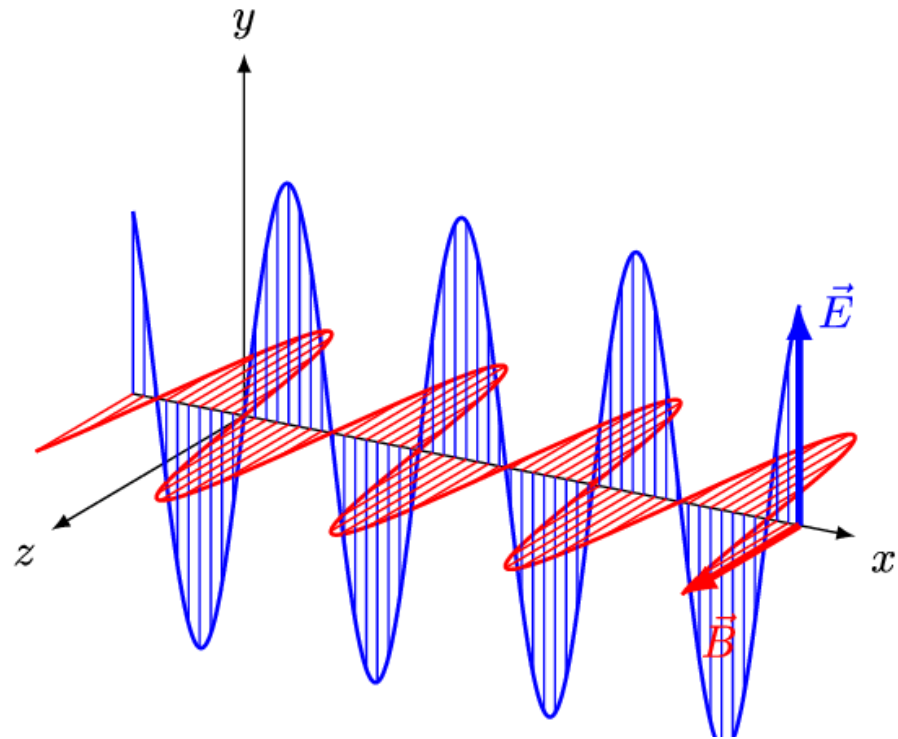
Longitudinal wave



Transverse Wave

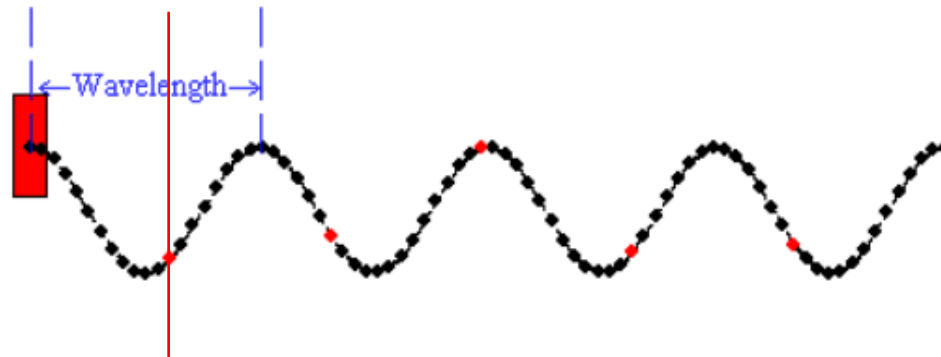


CHAPTER 3. WAVE

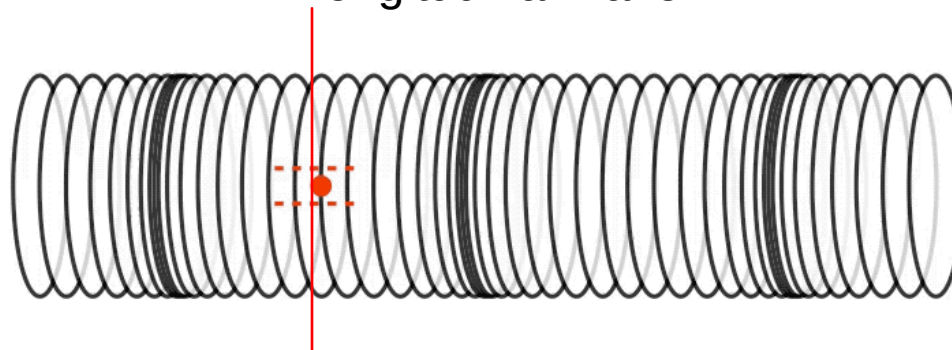


3.1 WAVE EQUATION-HARMONIC

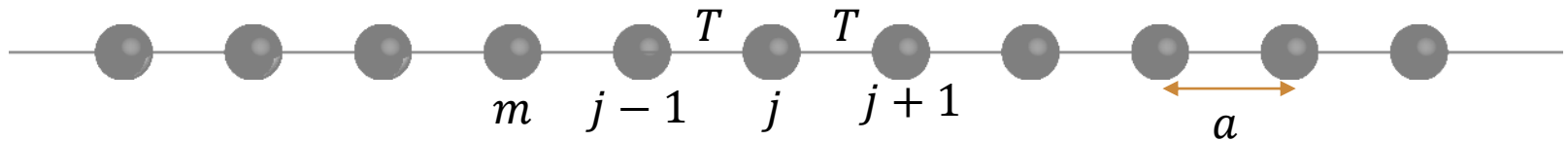
Transverse Wave



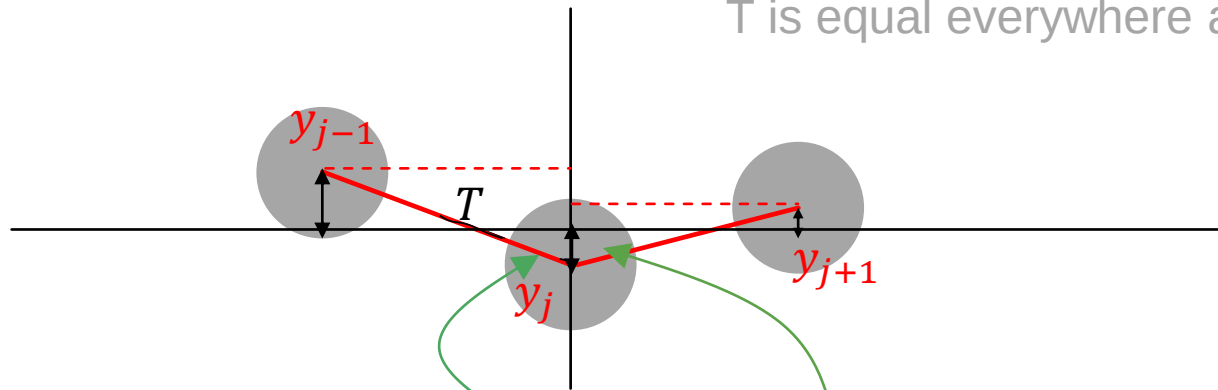
Longitudinal wave



3.1 WAVE EQUATION-HARMONIC



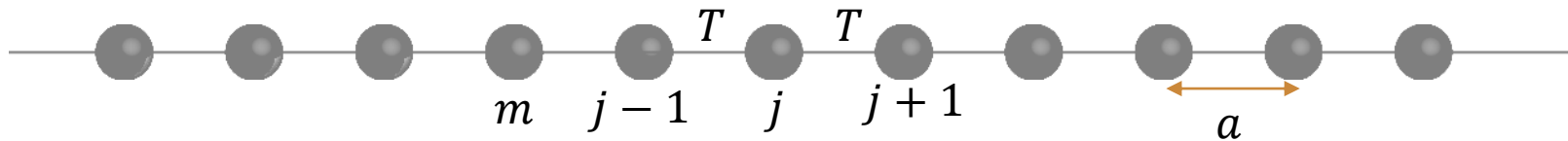
a massless string with tension T
 T is equal everywhere along the string



$$m\ddot{y}_j = -T \sin \theta_{j-1} - T \sin \theta_{j+1}$$

$$m\ddot{y}_j = -T \frac{y_j - y_{j-1}}{a} - T \frac{y_j - y_{j+1}}{a}$$

3.1 WAVE EQUATION-HARMONIC



$$m\ddot{y}_j = -T \frac{y_j - y_{j-1}}{a} - T \frac{y_j - y_{j+1}}{a}$$

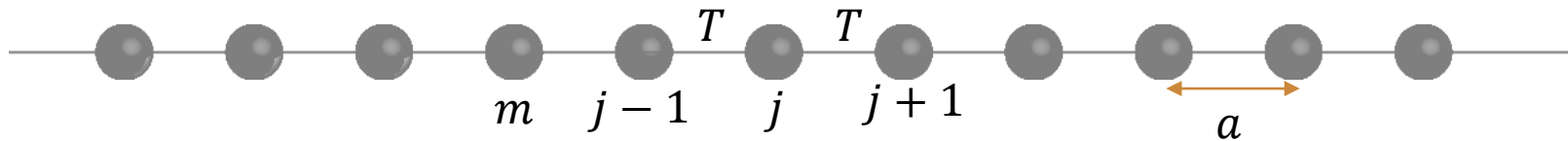
$$m\ddot{y}_j = T \frac{y_{j-1}}{a} - 2T \frac{y_j}{a} + T \frac{y_{j+1}}{a}$$

$$\ddot{y}_j = \frac{T}{ma} y_{j-1} - \frac{2T}{ma} y_j + \frac{T}{ma} y_{j+1}$$

$$\frac{T}{ma} = \omega_0^2$$

$$\ddot{y}_j = \omega_0^2 y_{j-1} - 2\omega_0^2 y_j + \omega_0^2 y_{j+1}$$

3.1 WAVE EQUATION-HARMONIC



⋮

$$\ddot{y}_{j-1} = \omega_0^2 y_{j-2} - 2\omega_0^2 y_{j-1} + \omega_0^2 y_j$$

$$\ddot{y}_j = \omega_0^2 y_{j-1} - 2\omega_0^2 y_j + \omega_0^2 y_{j+1}$$

$$\ddot{y}_{j+1} = \omega_0^2 y_j - 2\omega_0^2 y_{j+1} + \omega_0^2 y_{j+2}$$

⋮

$$\ddot{\mathbf{Y}} = -\mathbf{W}\mathbf{Y}$$

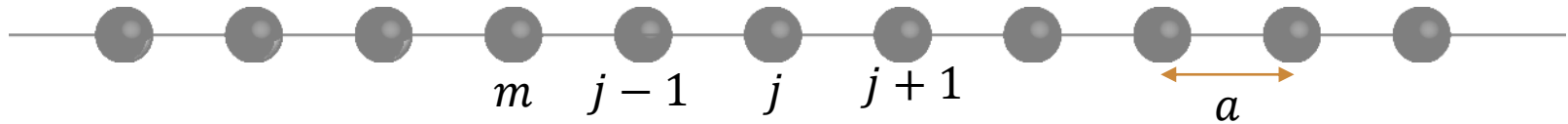
$$\mathbf{W} = \begin{pmatrix} \ddots & 0 & & 0 \\ 0 & 2\omega_0^2 & -\omega_0^2 & 0 \\ 0 & -\omega_0^2 & 2\omega_0^2 & -\omega_0^2 \\ 0 & 0 & -\omega_0^2 & 2\omega_0^2 \\ & & & \ddots \end{pmatrix}$$

W is symmetric

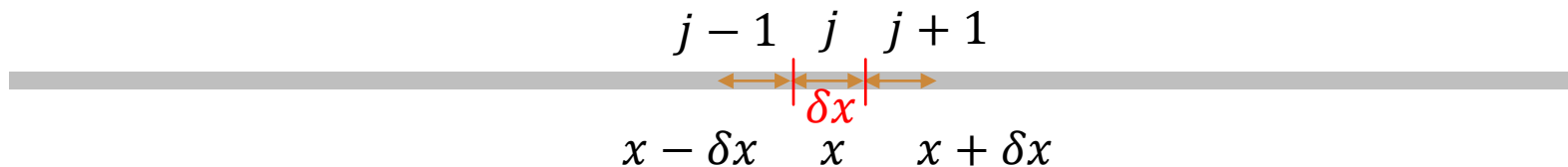
General solution: $\mathbf{Y} = \mathbf{A} \cos(\omega t + \varphi)$

of Normal mode ?

3.1 WAVE EQUATION-HARMONIC

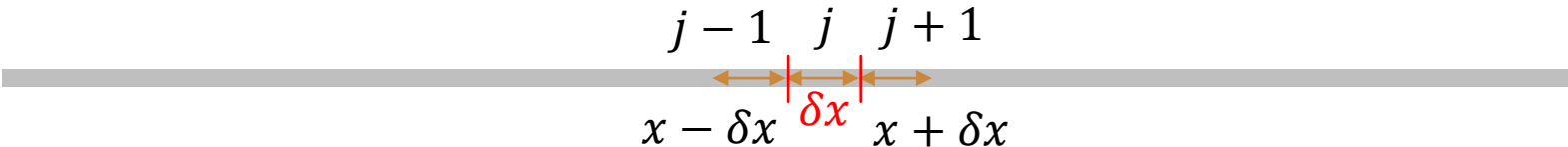


$$m\ddot{y}_j = -T \frac{y_j - y_{j-1}}{a} - T \frac{y_j - y_{j+1}}{a}$$



$$m\ddot{y}_j = -T \frac{y_j - y_{j-1}}{\delta x} + T \frac{y_{j+1} - y_j}{\delta x}$$

3.1 WAVE EQUATION-HARMONIC



$$\begin{array}{ccc}
 j-1 & j & j+1 \\
 \leftarrow & \text{red } \delta x & \rightarrow \\
 x - \delta x & & x + \delta x
 \end{array}$$

$$m\ddot{y}_j = -T \frac{y_j - y_{j-1}}{\delta x} + T \frac{y_{j+1} - y_j}{\delta x}$$

$$m = \rho \cdot \delta x \quad \text{线密度}$$

$$\ddot{y}_j = \left[-T \frac{\delta y}{\delta x} \Big|_{x=x_j} + T \frac{\delta y}{\delta x} \Big|_{x=x_{j+1}} \right] / (\rho \cdot \delta x) = \frac{T}{\rho} \left(y' \Big|_{x+\delta x} - y' \Big|_x \right) / \delta x = \frac{T}{\rho} y'' \Big|_x$$

$$\frac{\partial^2 y}{\partial t^2} \Big|_x = \frac{T}{\rho} \frac{\partial^2 y}{\partial x^2} \Big|_x$$

$$v_p^2 = \frac{T}{\rho} \rightarrow$$

$$\frac{\partial^2 y}{\partial t^2} = v_p^2 \frac{\partial^2 y}{\partial x^2}$$

wave equation
partial differential eq

3.2 W.E.-ANALYTICAL SOLUTION

Wave equation: $\frac{\partial^2 u}{\partial t^2} = v_p^2 \frac{\partial^2 u}{\partial x^2}$

$$\begin{aligned}\xi &= x + v_p t \\ \eta &= x - v_p t\end{aligned}$$

$$\frac{\partial u}{\partial t} = \frac{\partial u}{\partial \xi} \frac{\partial \xi}{\partial t} + \frac{\partial u}{\partial \eta} \frac{\partial \eta}{\partial t} = v_p \left(\frac{\partial u}{\partial \xi} - \frac{\partial u}{\partial \eta} \right)$$

$$\frac{\partial^2 u}{\partial t^2} = v_p^2 \left(\frac{\partial^2 u}{\partial \xi^2} - 2 \frac{\partial^2 u}{\partial \xi \partial \eta} + \frac{\partial^2 u}{\partial \eta^2} \right) \quad \frac{\partial^2 u}{\partial x^2} = \frac{\partial^2 u}{\partial \xi^2} + 2 \frac{\partial^2 u}{\partial \xi \partial \eta} + \frac{\partial^2 u}{\partial \eta^2}$$

→ $\frac{\partial^2 u}{\partial \xi \partial \eta} = 0 \quad u = f(\xi) + g(\eta)$

$$u = f(x + v_p t) + g(x - v_p t)$$

General solution

3.2 W.E.-ANALYTICAL SOLUTION

Wave equation:
$$\frac{\partial^2 u}{\partial t^2} = v_p^2 \frac{\partial^2 u}{\partial x^2}$$

In mathematics, a well-posed problem is one for which the following properties hold (定解问题的适定性):

- The problem has a solution (存在性)*
- The solution is unique (唯一性)*
- The solution's behavior changes continuously with the initial conditions (稳定性)*

3.2 W.E.- ANALYTICAL SOLUTION

1D Free Waves:
$$\frac{\partial^2 u}{\partial t^2} = v_p^2 \frac{\partial^2 u}{\partial x^2} \quad (-\infty < x < \infty, t > 0)$$

initial condition: $u(0, x) = \varphi(x); \quad \dot{u}(0, x) = \phi(x)$

In 1746, d'Alembert discovered the one-dimensional wave equation, and within ten years Euler discovered the three-dimensional wave equation.

general solution: $u(t, x) = f(x + v_p t) + g(x - v_p t)$

$$u(0, x) = f(x) + g(x) = \varphi(x) \quad \dot{f} = \frac{df(x + v_p t)}{d(x + v_p t)} \frac{d(x + v_p t)}{dt}$$

$$\dot{u}(0, x) = v_p f' - v_p g' = \phi(x)$$

$$f(x) - g(x) = \frac{1}{v_p} \int_0^x \phi(\xi) d(\xi) + c$$

3.2 W.E.- ANALYTICAL SOLUTION

1D Free Waves:
$$\frac{\partial^2 u}{\partial t^2} = v_p^2 \frac{\partial^2 u}{\partial x^2} \quad (-\infty < x < \infty, t > 0)$$

initial condition: $u(0, x) = \varphi(x); \quad \dot{u}(0, x) = \phi(x)$

$$f(x) = \frac{1}{2} \varphi(x) + \frac{1}{2v_p} \int_0^x \phi(\xi) d(\xi) + \frac{c}{2}$$

$$g(x) = \frac{1}{2} \varphi(x) - \frac{1}{2v_p} \int_0^x \phi(\xi) d(\xi) - \frac{c}{2}$$

$$u(t, x) = f(x + v_p t) + g(x - v_p t) = \frac{\varphi(x + v_p t) + \varphi(x - v_p t)}{2} + \frac{1}{2v_p} \int_{x-v_p t}^{x+v_p t} \phi(\xi) d\xi$$

d'Alembert's formula

It does not always work!

3.2 W.E.-ANALYTICAL SOLUTION

Wave equation: $\frac{\partial^2 u}{\partial t^2} = v_p^2 \frac{\partial^2 u}{\partial x^2}$ *Separable partial differential equation*



$$u(x, t) = \boxed{A(x)B(t)}$$

Spatial shape
Time evolution

$$\frac{\partial^2 u}{\partial t^2} = \frac{d^2 B(t)}{dt^2} A(x) \qquad \frac{\partial^2 u}{\partial x^2} = \frac{d^2 A(x)}{dx^2} B(t)$$

$$\frac{d^2 B(t)}{dt^2} A(x) = v_p^2 \frac{d^2 A(x)}{dx^2} B(t)$$

3.2 W.E.-ANALYTICAL SOLUTION

$$\frac{d^2 B(t)}{dt^2} A(x) = v_p^2 \frac{d^2 A(x)}{dx^2} B(t)$$

Separation of variables

$$\frac{d^2 B(t)}{v_p^2 B(t) dt^2} = \frac{d^2 A(x)}{A(x) dx^2} = -k_m^2 \quad k_m \text{ is a constant}$$

$$\frac{d^2 B(t)}{v_p^2 B(t) dt^2} = -k_m^2 \quad \frac{d^2 B(t)}{dt^2} = -k_m^2 v_p^2 B(t)$$

$$\ddot{B} = -\omega_m^2 B(t)$$

$$B(t) = \beta \sin(\omega_m t + \phi) \quad \omega_m^2 = k_m^2 v_p^2$$

3.2 W.E.- ANALYTICAL SOLUTION

$$\frac{d^2 B(t)}{dt^2} A(x) = v_p^2 \frac{d^2 A(x)}{dx^2} B(t)$$

Separation of variables

$$\frac{d^2 B(t)}{v_p^2 B(t) dt^2} = \frac{d^2 A(x)}{A(x) dx^2} = -k_m^2$$

$$\frac{d^2 A(x)}{A(x) dx^2} = -k_m^2 \quad \frac{d^2 A(x)}{dx^2} = -k_m^2 A(x)$$

$$A(x) = \alpha \sin(k_m x + \varphi)$$

3.2 W.E.- ANALYTICAL SOLUTION

$$u(x, t) = A(x)B(t)$$

$$u_m(x, t) = c_m \sin(k_m x + \varphi) \sin(\omega_m t + \phi)$$

$$u_m(x_1, t) = c_m \sin(k_m x_1 + \varphi) \sin(\omega_m t + \phi)$$

Normal Modes !

$$u_m(x_2, t) = c_m \sin(k_m x_2 + \varphi) \sin(\omega_m t + \phi)$$

Any two points oscillate in the same frequency with a fixed phase difference.

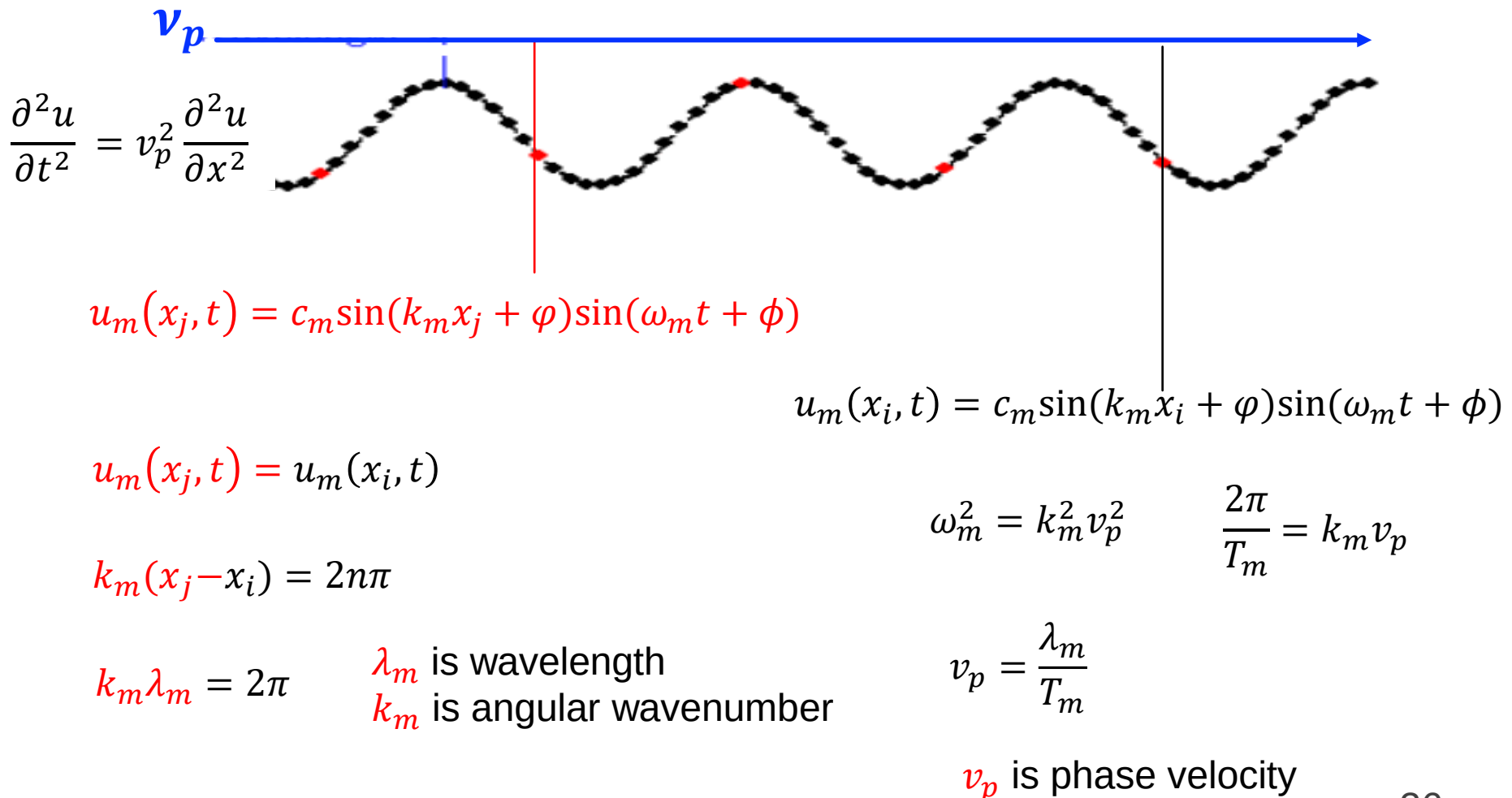
m th normal mode for continuous wave equation

$$u_m(x, t) = c_m \sin(k_m x + \varphi) \sin(\omega_m t + \phi)$$

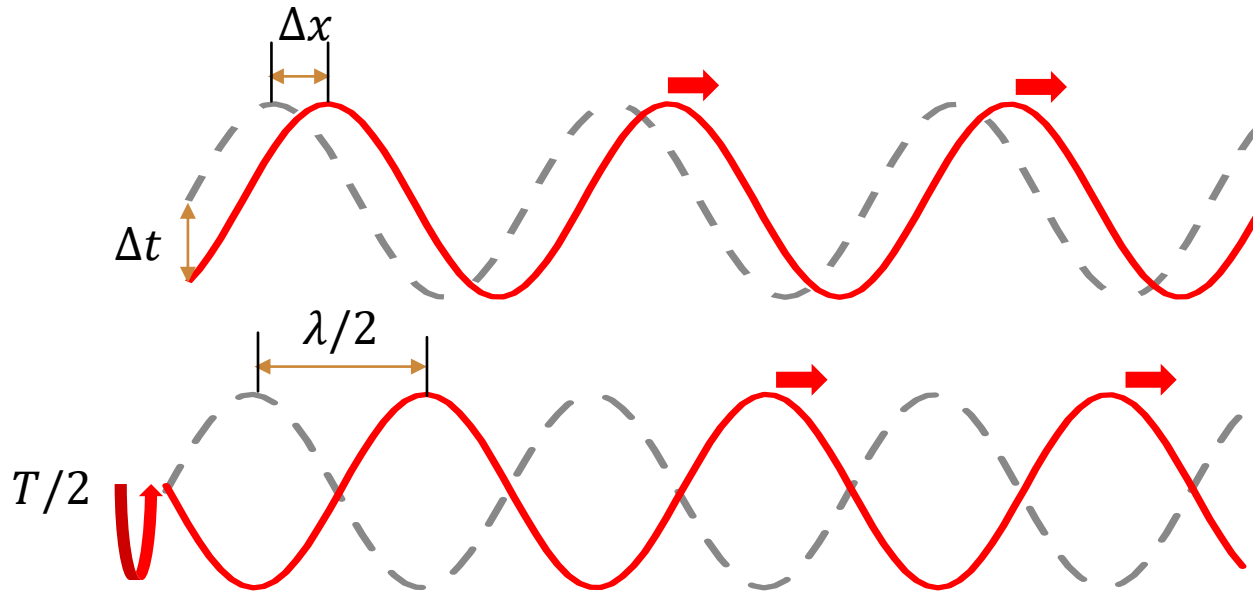
General solution

$$u(x, t) = \sum_m c_m \sin(k_m x + \varphi) \sin(\omega_m t + \phi)$$

3.2 W.E.- ANALYTICAL SOLUTION



3.2 W.E.- ANALYTICAL SOLUTION



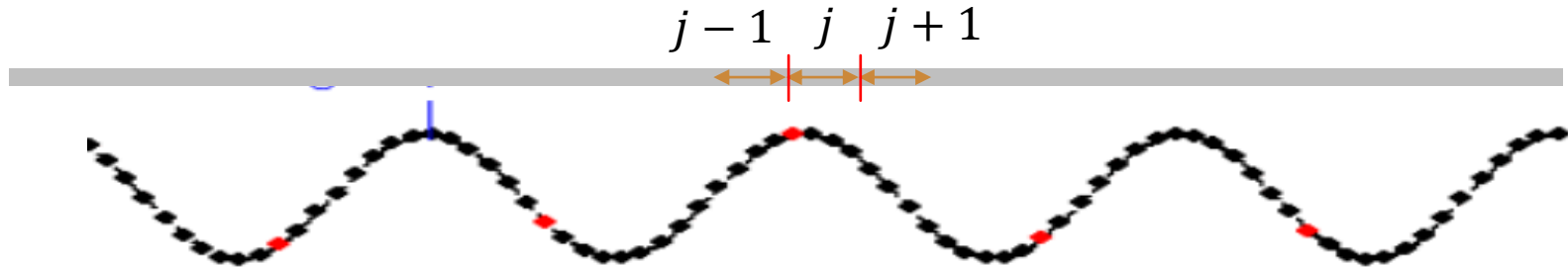
$$v_p = \frac{\Delta x}{\Delta t}$$

$$v_p = \frac{\Delta x}{\Delta t} = \frac{\lambda_m}{T_m} = \frac{\omega_m}{k_m}$$

$$k_m \lambda_m = 2\pi$$

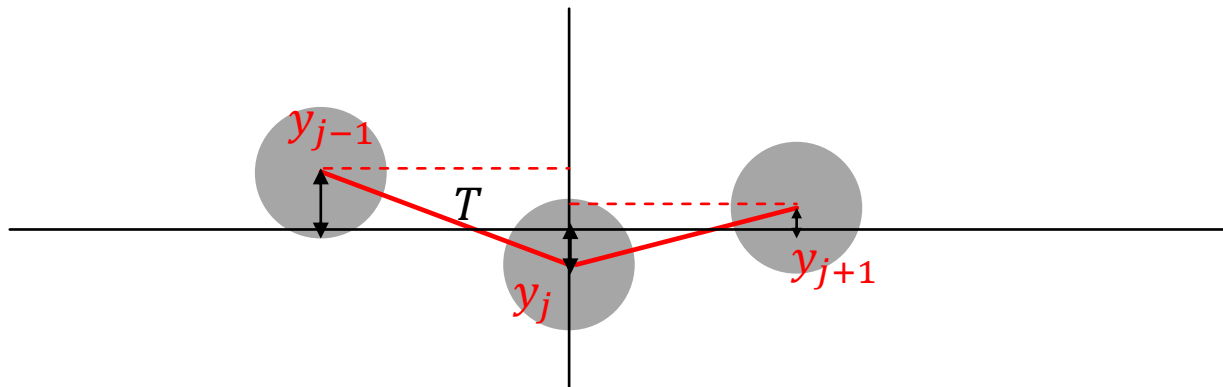
Dispersion relation: $v_p = \frac{\omega}{k}$

3.2 W.E.- ANALYTICAL SOLUTION

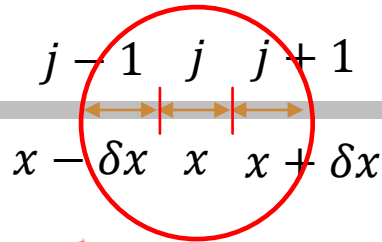


$$\left. \frac{\partial^2 y}{\partial t^2} \right|_x = \frac{T}{\rho} \left. \frac{\partial^2 y}{\partial x^2} \right|_x \quad \xrightarrow{v_p^2 = \frac{T}{\rho}} \quad \frac{\partial^2 y}{\partial t^2} = v_p^2 \frac{\partial^2 y}{\partial x^2} \quad \text{wave equation}$$

$$v_p = \sqrt{\frac{T}{\rho}}$$



3.2 W.E.- ANALYTICAL SOLUTION



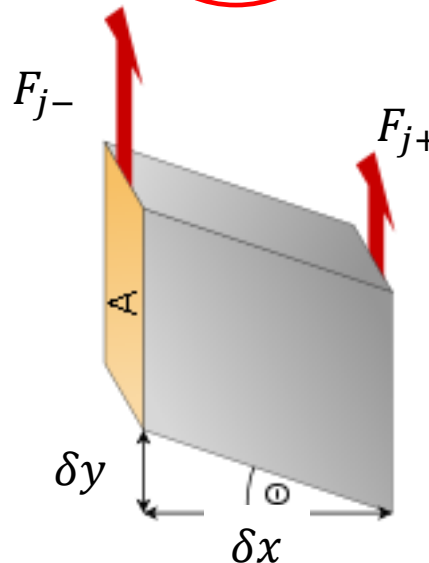
G : shear modulus

$$G = -\frac{F/A}{\delta y/\delta x} \quad \frac{\text{shear stress}}{\text{shear strain}}$$

$$F_{j-} = -\frac{AG}{\delta x}(y_j - y_{j-1})$$

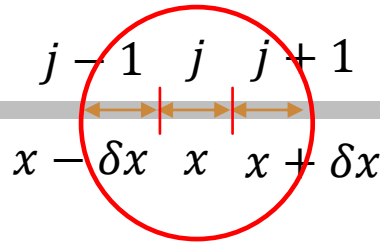
$$F_{j+} = -\frac{AG}{\delta x}(y_j - y_{j+1})$$

$$\delta m \ddot{y}_j = F_{j-} + F_{j+}$$



$$\delta m \ddot{y}_j = -AG \left. \frac{\delta y}{\delta x} \right|_{x=x_j} + AG \left. \frac{\delta y}{\delta x} \right|_{x=x_j+\delta x}$$

3.2 W.E.- ANALYTICAL SOLUTION



$$\delta m \ddot{y}_j = -AG \left. \frac{\delta y}{\delta x} \right|_{x=x_j} + AG \left. \frac{\delta y}{\delta x} \right|_{x=x_j+\delta x}$$

$$\delta m \ddot{y}_j = AG (y'_{x=x_j+\delta x} - y'_{x=x_j}) / \delta x \cdot \delta x = AG \left. \frac{\partial^2 y}{\partial x^2} \right|_{x=x_j} \delta x$$

$$\ddot{y}_j = \frac{G}{\rho} \left. \frac{\partial^2 y}{\partial x^2} \right|_{x=x_j}$$

$$\frac{\partial^2 y}{\partial t^2} = v_p^2 \frac{\partial^2 y}{\partial x^2}$$

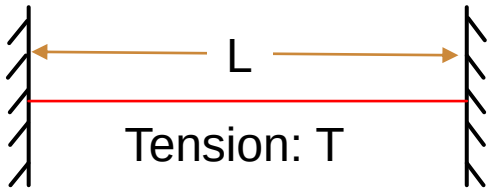
$$v_p = \sqrt{\frac{G}{\rho}}$$

3.2 BOUNDARY CONDITIONS

m th normal mode for continuous wave equation

$$u_m(x, t) = c_m \sin(k_m x + \varphi) \sin(\omega_m t + \phi)$$

In specific cases: Two fix ends



$$u_m(0, t) = c_m \sin(\varphi) \sin(\omega_m t + \phi) = 0$$

$$\varphi = n\pi$$

$$u_m(L, t) = c_m \sin(k_m L + \varphi) \sin(\omega_m t + \phi) = 0$$

Boundary conditions:

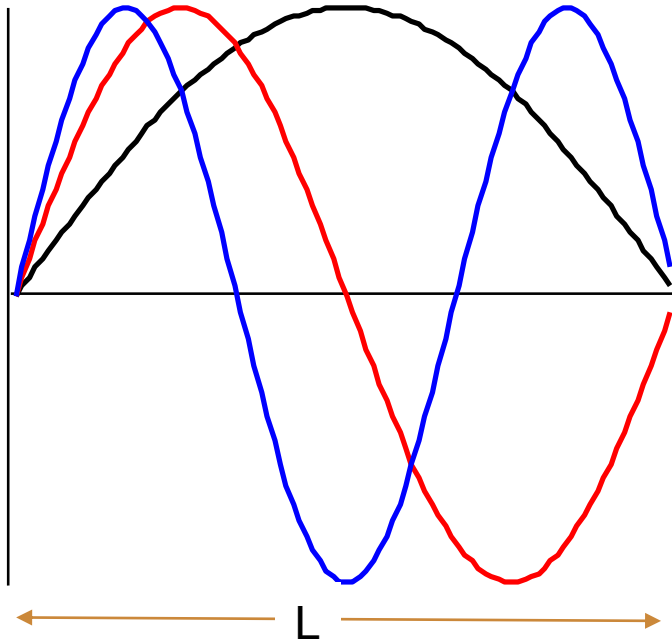
$$u(0, t) = 0$$

$$u(L, t) = 0$$

$$k_m L + \varphi = n'\pi$$

$$k_m = m\pi/L$$

3.2 BOUNDARY CONDITIONS



$$k_m = \frac{m\pi}{L}$$

$$\lambda_m = \frac{2\pi}{k_m} = \frac{2L}{m}$$

$$\lambda_1 = 2L \quad k_1 = \frac{\pi}{L}$$

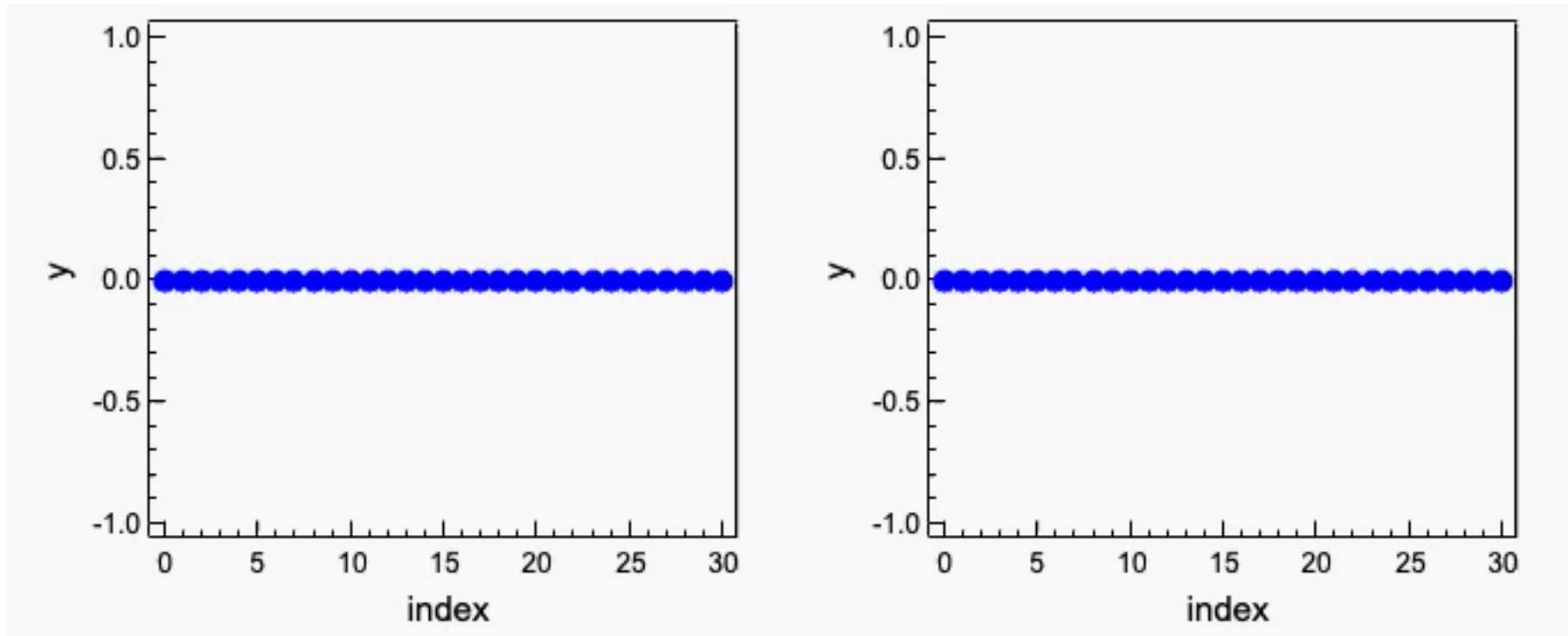
$$\lambda_2 = L \quad k_2 = \frac{2\pi}{L}$$

$$\lambda_3 = \frac{2}{3}L \quad k_3 = \frac{3\pi}{L}$$

$$\lambda_4 = \frac{1}{2}L \quad k_4 = \frac{4\pi}{L}$$

3.2 BOUNDARY CONDITIONS

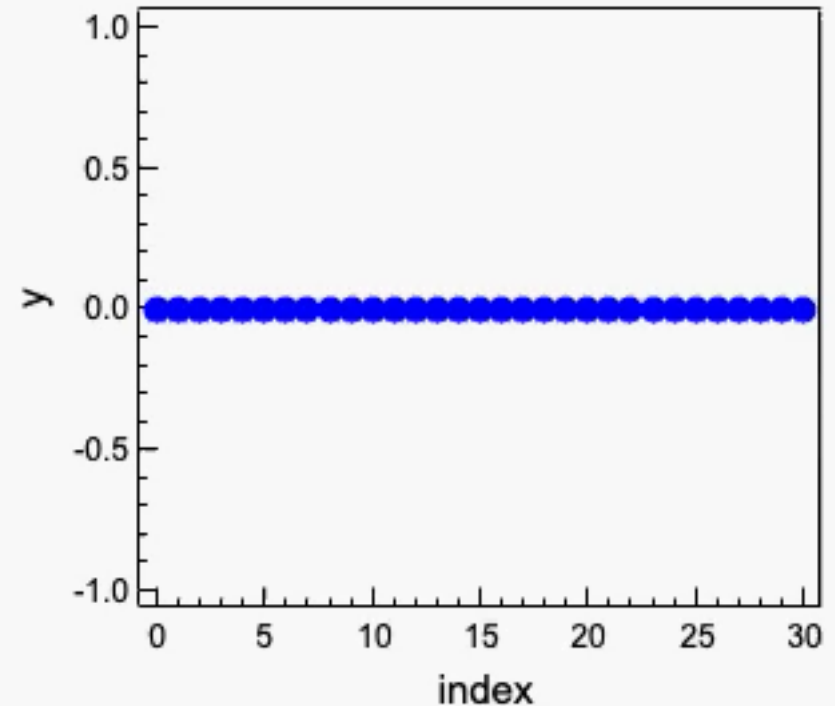
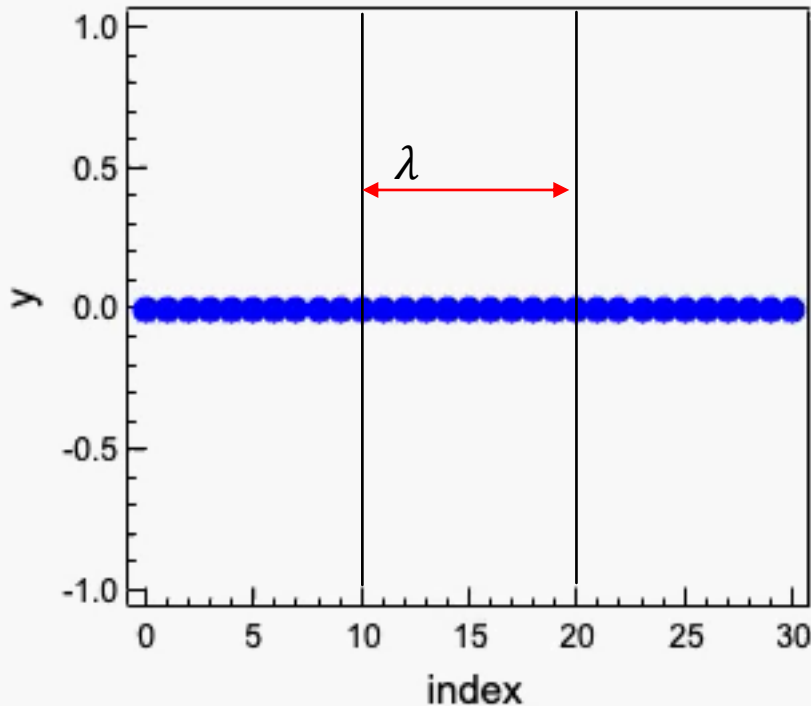
Normal modes



$$\lambda_1 = 2L \quad k_1 = \frac{\pi}{L} \quad \frac{\omega_m}{k_m}$$

$$\lambda_2 = L \quad k_2 = \frac{2\pi}{L}$$

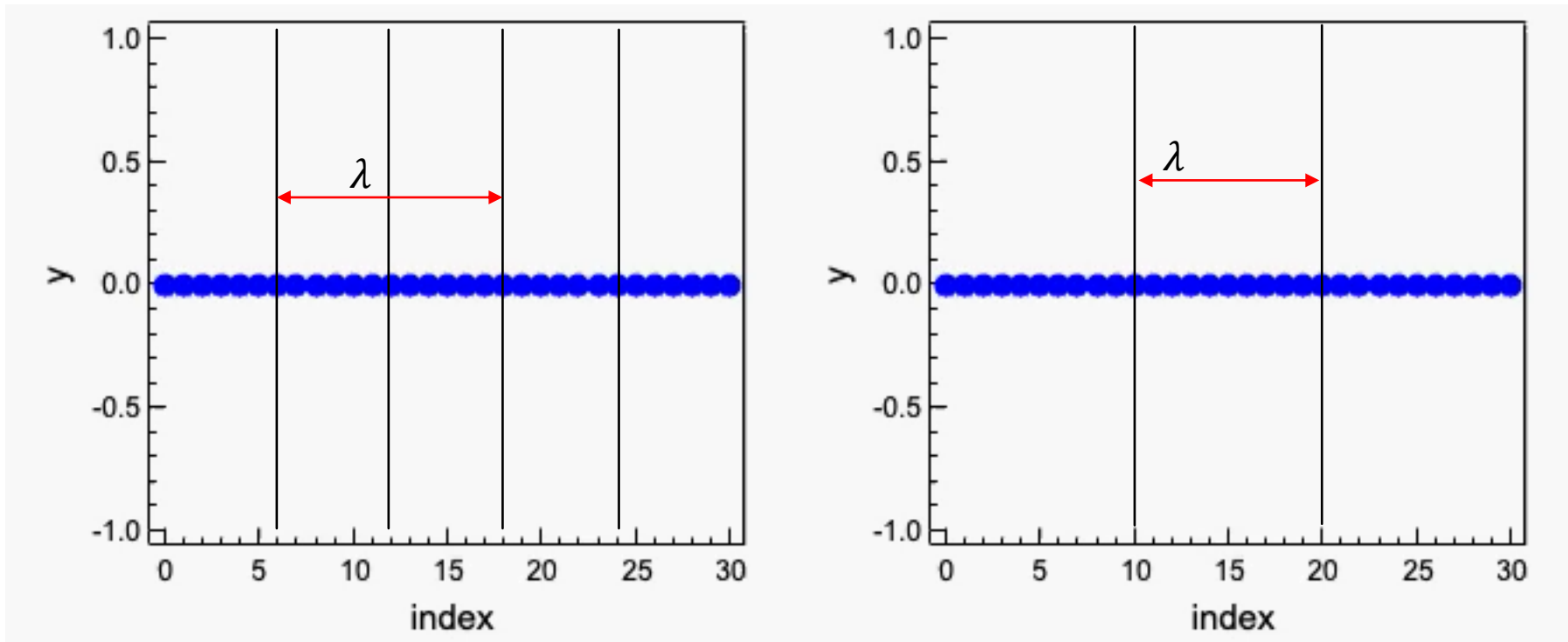
3.2 BOUNDARY CONDITIONS



$$\lambda_3 = \frac{2}{3}L \quad k_3 = \frac{3\pi}{L}$$

$$\lambda_4 = \frac{1}{2}L \quad k_4 = \frac{4\pi}{L}$$

3.2 BOUNDARY CONDITIONS



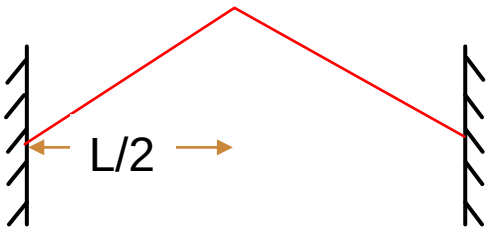
$$\lambda_5 = \frac{2}{5}L \quad k_5 = \frac{5\pi}{L}$$

$$\omega_m = v_p k_m$$

$$\lambda_3 = \frac{2}{3}L \quad k_3 = \frac{3\pi}{L}$$

k is wavenumber, measuring the spatial frequency of a wave

NEXT LECTURE



Initial condition $t=0$:

General solution

$$u(x, t) = \sum_m c_m \sin(k_m x + \phi) \sin(\omega_m t + \phi)$$

Boundary conditions: $k_m = \frac{m\pi}{L}$

Initial condition

$$u(x, 0) = \sum_m c_m \sin(k_m x + \phi) \sin(\phi)$$

Can $\sum_m c_m \sin(k_m x + \phi) \sin(\phi)$ construct $u(x, 0)$?